

Linear Regression

Tuesday, 1 April 2025

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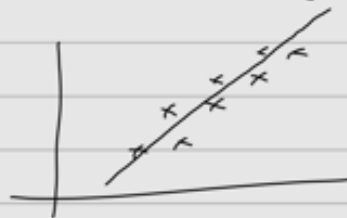


Linear regression...

* Linear Regression OLS :-

↳ OLS $\Rightarrow$ ordinary least squares.

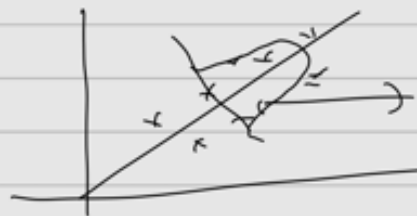
$\Rightarrow$ Assumption there is there exist a line that roughly model the data (not classify it)



$$\epsilon_i \text{ is error} \\ = \underline{\underline{y_i - w^T x_i}}$$

i.e $y_i = w^T x_i + \epsilon_i$, where ϵ_i is some noise drawn from the gaussian distribution $\epsilon_i \sim N(0, \sigma^2)$.

* $\hookrightarrow$ This gaussian distribution lies on the vertical axis



Signifies that point are concentrated on the line by having most frequency on line.

$$p(y_i | \vec{x}_i; w) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(w^T x_i - y_i)^2}{2\sigma^2}}$$

$$MLE = \underset{w}{\operatorname{argmax}} \prod_{i=1}^n p(y_i | \vec{x}_i; w)$$

$$= \operatorname{argmax}_w \sum_{i=1}^n \log P(y_i | x_i^T w)$$

$$= \operatorname{argmax}_w \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2\sigma^2} (w^T x_i - y_i)^2}$$

$$= \operatorname{argmin}_w \sum_{i=1}^n (w^T x_i - y_i)^2 = \frac{1}{n} \operatorname{argmin}_w \sum_{i=1}^n (w^T x_i - y_i)^2$$

MAP:-

$$P(w | y_1, x_1, \dots, y_n, x_n) = \frac{P(D|w) \cdot P(w)}{Z}$$

$\Rightarrow$ Map assumes w follows some gaussian distribution.

$$= \operatorname{argmax}_w P(w | y_1, x_1, y_2, x_2, \dots, y_n, x_n)$$

$$= \operatorname{argmax}_w P(y_1, x_1, \dots, y_n, x_n | w) \cdot P(w)$$

$$= \operatorname{argmax}_w \prod_i P(y_i | x_i, w) P(w)$$

$\hookrightarrow$ Because each y_i depends only on each x_i
 $\hookrightarrow$ as y_i 's are independent.
 (Not an assumption).

$$= \operatorname{argmax}_w \sum_i \log (P(y_i | x_i, w) \cdot P(w)), \quad P(w) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{w^T w}{2\sigma^2}}$$

$$\operatorname{argmax}_w \frac{1}{2\sigma^2} \sum_{i=1}^n (w^T x_i - y_i)^2 + \frac{1}{2\sigma^2} w^T w$$

$$= \frac{1}{N} \left(\sum_{i=1}^n (w^T x_i - y_i)^2 + \lambda \|w\|_2^2 \right)$$

$$\text{where } \lambda = \frac{\sigma^2}{t^2}$$

MAP
expression

$$J(w) = \frac{1}{n} \sum_{i=1}^n (w^T x_i - y_i)^2 + \lambda \|w\|^2$$

$$x^T = [x_1 \dots x_n], \quad y^T = [y_1 \dots y_n].$$

$$(xw - y)^2 = (xw - y)^T (xw - y).$$

$$J(w) = w^T x^T x w - 2 y^T x w + y^T y$$

$$\frac{dJ(w)}{dw} = 0 \quad \text{gives minimum (convex fn).}$$

$$\therefore x^T x w = x^T y$$

$$w = (x^T x)^{-1} x^T y.$$

gives closed form solution.

↳ very costly to compute.